GOVT OF INDIA
HYDROVISION AI
NDSA • CWC
STATUTORY SYSTEM

[redacted] • GOVERNMENT OF INDIA

[redacted] • MINISTRY OF JAL SHAKTI

[redacted] • NATIONAL DAM SAFETY AUTHORITY

HYDROVISION AI

AI-Assisted Dam Inspection & Predictive Maintenance Decision Support System

COMPLETE FINAL PROJECT BLUEPRINT & TECHNICAL SPECIFICATIONS

A comprehensive top-to-bottom architecture reference, database models breakdown, AI detection pipeline, 3D Digital Twin engine, GIS spatial analysis, risk scoring algorithms, automated PDF generation engine, RBAC security design, and deployment protocol.

Project Code:	HYDROVISION-AI-2026-FINAL	**Target Framework:**	Dam Safety Act 2021 / NDSA / CWC
Backend Stack:	Python FastAPI, SQLAlchemy 2.0, ReportLab	**Frontend Stack:**	React 18, TypeScript, Three.js, Leaflet
AI Engines:	YOLO11, OpenCV, XGBoost Regressor	**Spatial Database:**	PostgreSQL 16 + PostGIS (SQLite Fallback)
Document Version:	v1.0.0 (Production Release)	**Publication Date:**	September 29, 2026
System Status:	**OPERATIONAL & CERTIFIED**	**Classification:**	Official Statutory Government Document

CRYPTOGRAPHIC INTEGRITY STAMP & SYSTEM SIGNATURE
SHA-256 Checksum: `248ae9c41388819be8333e297aef2e2387ecc9127bf58569b40055433abe5fc7`
Verification Protocol: QR Scan / Automated API Auth
This document provides complete top-to-bottom technical details of the HydroVision AI platform.

1. EXECUTIVE SUMMARY & SYSTEM OVERVIEW

HydroVision AI is a state-of-the-art, government-grade infrastructure decision support platform engineered specifically for large dams, hydroelectric power stations, spillways, and water storage structures. Modern dam safety management under India's **Dam Safety Act 2021** and guidelines established by the **National Dam Safety Authority (NDSA)** and **Central Water Commission (CWC)** demands rapid, repeatable, and non-destructive inspection methodologies. HydroVision AI addresses these needs by combining high-resolution drone imagery, computer vision, AI deep learning, 3D Digital Twin visualization, spatial GIS indexing, condition scoring algorithms, risk evaluation matrices, and automated PDF reporting into a closed-loop operational workflow.

Key Objectives & Core Mission:

- **Automated AI Defect Detection:** Replace subjective visual inspections with YOLO11 deep learning object detection capable of identifying structural cracks, spalling, seepage/efflorescence, rebar corrosion, concrete erosion, and vegetation growth.
- **Quantitative Computer Vision Measurements:** Utilize OpenCV spatial transformation algorithms to convert pixel-level defect contours into precise millimeter measurements (length, width, surface area) based on camera GSD (Ground Sample Distance).
- **Human-in-the-Loop Engineering Control:** Provide inspection engineers with dedicated UI tools to review, verify, adjust, or reject AI-detected anomalies prior to statutory logging.
- **Interactive 3D Digital Twin & GIS Mapping:** Project verified defects onto photogrammetric 3D dam meshes (GLTF/GLB) via Three.js WebGL rendering and PostGIS spatial coordinates.
- **Predictive Deterioration & Risk Engine:** Compute Dam Health Index (DHI), execute Failure Mode and Effects Analysis (FMEA), and project Remaining Service Life (RSL) using XGBoost trend regression.
- **Closed-Loop Maintenance Lifecycle:** Seamlessly bridge defect detection with risk-prioritized maintenance work orders, contractor assignment, repair evidence verification, and engineer reinspection closure.
- **Statutory ReportLab PDF Generation:** Automatically compile tamper-evident, QR-verified PDF audit reports with dual-language headers and SHA-256 cryptographic signatures.

2. SYSTEM ARCHITECTURE & TECH STACK

HydroVision AI follows a modern, decoupled, microservice-ready modular architecture consisting of a high-performance Python FastAPI backend, a responsive React 18 TypeScript frontend, a PostgreSQL/PostGIS spatial database, MinIO object storage, and a deep learning computer vision pipeline.

Layer / Component	Technologies & Frameworks	Key Responsibility & Function
Frontend UI	React 18, TypeScript, Vite, Tailwind CSS	5 Core Hub SPA interface, responsive styling, RBAC guards, state management
3D Visualization	Three.js, WebGL, GLTFLoader, OrbitControls	Interactive 3D Digital Twin rendering, raycasting defect marker placement, spatial heatmaps
GIS & Mapping	Leaflet, React-Leaflet, OpenStreetMap	Geographic dam markers, spatial defect overlays, satellite view toggle, coordinate conversion
Backend API	Python 3.11, FastAPI, Pydantic v2, Uvicorn	Asynchronous RESTful APIs, request validation, business logic, dependency injection
Database Layer	PostgreSQL 16 + PostGIS / SQLite, SQLAlchemy 2.0	Relational data modeling, spatial indexing (GiST), transactional integrity, migrations
AI Deep Learning	YOLO11 (Ultralytics), PyTorch, CUDA	Defect bounding box localization, classification, confidence scoring, batch processing

Computer Vision	OpenCV (cv2), NumPy, Pillow	EXIF extraction, image quality scoring (blur/brightness), pixel-to-mm scaling, skeletonization
Predictive Engine	XGBoost, Scikit-learn, Pandas	Historical defect trajectory analysis, growth rate velocity, Remaining Service Life (RSL) forecast
Reporting Engine	ReportLab 4.x, PyPDF, QrCode	Dynamic PDF publishing, vector seal rendering, QR code generation, cryptographic signing
Security & Auth	JWT (JSON Web Tokens), Passlib (Bcrypt)	Role-Based Access Control (RBAC), password hashing, stateless session validation

3. ROLE-BASED ACCESS CONTROL (RBAC) & PORTALS

HydroVision AI enforces strict Role-Based Access Control (RBAC) tailored to official government hierarchy and operational division of duties. The system provides 5 primary portals plus public grievance access:

User Role	Target Persona	Portal Capabilities & Scope
AUTHORITY	Chief Engineer / NDSA Official / Ministry Executive	Macro multi-dam dashboard, state-wide risk matrix, compliance summary, audit logs, public grievance oversight, strategic directive review.
DAM OPERATOR	Site Dam Manager / Operations Supervisor	Assigned dam monitoring, real-time reservoir & water level telemetry logs, local alert monitoring, operational history access.
INSPECTION ENGINEER	Drone Pilot / Structural Inspection Engineer	Drone dataset upload, image quality validation, AI detection triggering, defect review/editing, condition index calculation, PDF report generation.
MAINTENANCE MANAGER	Executive Repair Engineer / Maintenance Lead	Risk-prioritized work order creation, repair team assignment, execution progress tracking, repair evidence upload, reinspection requests.
SUPREME COMMAND	High-Level Ministry Authority / Emergency Council	Emergency command override, supreme rating assignment (A+ to D), direct binding directive issuance across all state dams.
PUBLIC CITIZEN	Local Resident / Public Reporter	Public Grievance submission (seepage, cracks, flooding risk), GPS pin selection, photo upload, tracking code status lookup.

4. END-TO-END 14-STAGE OPERATIONAL LIFECYCLE

HydroVision AI enforces a complete 14-stage closed-loop operational workflow from raw data acquisition to final maintenance closure:

Stage & Title	Description & Execution Detail
1. DRONE CAPTURE	High-resolution flight capture across dam structure (Upstream face, Spillway, Downstream face, Crest, Gallery).
2. DATA UPLOAD	Batch ZIP / multi-file imagery ingestion via Inspection Engineer portal with flight metadata.
3. DATA VALIDATION	Automated EXIF extraction, GPS verification, blur score calculation, and resolution compliance check.
4. AI DETECTION	YOLO11 model execution to identify bounding boxes for cracks, spalling, seepage, and corrosion.
5. CV MEASUREMENT	OpenCV pixel-to-millimeter GSD scaling to calculate crack length (mm), width (mm), and area (sq cm).
6. ENGINEER VERIFICATION	Human-in-the-loop review: engineer accepts, modifies bounding boxes, adjusts severity, or rejects false positives.
7. GEO-LOCATION MAPPING	Spatial PostGIS coordinate calculation and zone tag assignment (e.g. Zone 2 - Spillway Pier 3).
8. 3D DIGITAL TWIN	Projecting verified defect pins onto photogrammetric 3D GLTF dam model via Three.js raycasting.
9. HISTORICAL COMPARISON	Spatial matching against prior inspection datasets to compute defect growth rate (delta mm/year).
10. CONDITION ASSESSMENT	Computation of Dam Health Index (DHI) using weighted zone and severity metrics.
11. RISK EVALUATION	FMEA risk matrix calculation evaluating Likelihood vs Consequence of Failure.
12. PREDICTIVE ANALYSIS	XGBoost regression forecasting Remaining Service Life (RSL) and critical threshold breach dates.
13. MAINTENANCE ACTION	Risk-driven work order generation, repair team dispatch, and post-repair evidence upload.
14. REINSPECTION & CLOSURE	Engineer reinspection verification, statutory ReportLab PDF compilation, and formal work order closure.

5. DATABASE SCHEMA & DATA MODELS BREAKDOWN

The relational schema is defined using SQLAlchemy 2.0 with spatial PostGIS types (`Geometry('POINT', 4326)`) and standard SQLite fallback decorators. Below is a summary of the 17 primary database entities:

Model Class	Table Name	Key Attributes / Fields	Business Purpose
User	`users`	`id, email, hashed_password, full_name, role (Enum), department, is_active, created_at`	System user accounts with RBAC roles (AUTHORITY, DAM_OPERATOR, INSPECTION_ENGINEER, MAINTENANCE_MANAGER, SUPREME_COMMAND).
Dam	`dams`	`id, name, code, state, district, dam_type, dam_category, height_meters, length_meters, storage_capacity_mcm, latitude, longitude, geometry, status, supreme_rating`	Master registry of dam assets, geographical coordinates, physical dimensions, and operational status.
SupremeCommandDirective	`supreme_command_directives`	`id, dam_id, directive_code, title, command_text, priority, status, issued_by_id, target_completion_date`	Binding strategic directives issued by high-level authorities to dam operating teams.
DamZone	`dam_zones`	`id, dam_id, zone_name, criticality, description`	Structural zone segmentation (Upstream Face, Downstream Face, Spillway, Crest, Gallery, Stilling Basin).
Inspection	`inspections`	`id, inspection_code, dam_id, zone_id, inspection_type, assigned_engineer_id, inspection_date, status (Enum), coverage_percentage`	Inspection sessions tracking coverage, flight date, assigned engineer, and workflow stage.
DroneDataset	`drone_datasets`	`id, inspection_id, dataset_name, drone_model, total_files, valid_files, warning_files, rejected_files, upload_status`	Batch drone imagery container holding dataset statistics and validation metrics.
DroneImage	`drone_images`	`id, dataset_id, inspection_id, dam_id, filename, file_path, file_size_bytes, file_hash_sha256, blur_score, brightness_score, latitude, longitude, altitude, quality_status`	Individual drone captured photos with EXIF metadata, GPS coordinates, SHA-256 hash, and quality metrics.
AIJob	`ai_jobs`	`id, job_code, inspection_id, media_id, job_type, status, model_name, total_media, eligible_media, started_at, completed_at`	Asynchronous AI batch execution job tracking execution time and model specifications.
AIDetection	`ai_detections`	`id, detection_code, ai_job_id, inspection_id, media_id, defect_type, confidence, bbox_x_min/y_min/x_max/y_max, review_status, final_defect_type, final_severity`	Raw AI-generated bounding boxes preserved with original vs engineer-edited decision fields.
Defect	`defects`	`id, defect_code, dam_id, inspection_id, zone_id, image_id, source_detection_id, defect_type, severity (Enum), estimated_length_mm, estimated_width_mm, pos_x_3d/y_3d/z_3d, status (Enum)`	Engineer-verified structural defects with millimeter measurements, 3D spatial positions, and lifecycle status.
DefectHistory	`defect_history`	`id, defect_id, inspection_id, severity, estimated_length_mm, status, change_trend, recorded_at`	Temporal defect state snapshot logging historical growth and severity changes across inspections.

DefectComparison	defect_comparisons	id, dam_id, current_inspection_id, historical_inspection_id, current_defect_id, historical_defect_id, match_method, match_distance_meters, comparison_status, measurement_change_mm, deterioration_status	Delta matching model comparing current defects against previous inspections to calculate growth velocity.
RiskAssessment	risk_assessments	id, dam_id, inspection_id, risk_score, risk_level, condition_index, condition_rating, contributing_factors, assessed_at	Calculated risk scores (0-100), health indices (0-100), rating tags, and contributing factor lists.
MaintenanceWorkOrder	maintenance_work_orders	id, work_order_code, dam_id, defect_id, assigned_manager_id, priority, assigned_team, description, recommended_action, due_date, status (Enum), repair_evidence_file	Risk-driven repair work orders tracking contractor teams, due dates, and post-repair proof.
Reinspection	reinspections	id, work_order_id, engineer_id, requested_at, completed_at, result_status, findings_notes	Post-maintenance verification record confirming repair quality prior to work order closure.
Report	reports	id, report_code, dam_id, inspection_id, generated_by_id, report_title, file_path, generated_at	Statutory PDF report metadata linking generated files to dam assets and generating users.
AuditLog	audit_logs	id, user_id, user_email, action, entity_type, entity_id, old_value, new_value, timestamp	Immutable security log capturing every user action, login, verification, and PDF generation event.
DigitalTwinModel	digital_twin_models	id, dam_id, name, file_name, storage_key, model_format, version, is_active, uploaded_by_id	Registry of 3D GLTF/GLB photogrammetric mesh models linked to specific dam assets.
PublicComplaint	public_complaints	id, tracking_code, citizen_name, citizen_phone, dam_id, category, title, description, latitude, longitude, ai_risk_score, status, work_order_id	Public grievance submissions with GPS location, AI risk classification, and resolution tracking.

6. BACKEND API SPECIFICATIONS & MODULE BREAKDOWN

The backend API is structured into modular FastAPI routers mounted under the `/api/v1` namespace. Below is an exhaustive reference of all API router domains:

Router Domain	Key Endpoints	Functional Description
Auth (`/api/v1/auth`)	`POST /login, POST /register, GET /me`	JWT token issue (OAuth2 bearer), user registration, current user profile retrieval.
Dams (`/api/v1/dams`)	`GET /, POST /, GET /{id}, GET /{id}/summary`	Dam asset CRUD, state/district filtering, summary health metrics, zone listing.
Inspections (`/api/v1/inspections`)	`GET /, POST /, GET /{id}, PUT /{id}/status`	Inspection session scheduling, flight status transition, coverage metrics.
Uploads (`/api/v1/uploads`)	`POST /images, POST /dataset, GET /quality-check`	Drone image upload handling, dataset ZIP unzipping, EXIF GPS parsing, quality validation.
AI (`/api/v1/ai`)	`POST /process/{inspection_id}, GET /detections/{inspection_id}`	Trigger YOLO11 defect detection on inspection dataset, return raw bounding boxes.
Defects (`/api/v1/defects`)	`GET /, PUT /{id}/verify, POST /{id}/spatial`	Defect listing, engineer verification/editing, millimeter measurement updates, 3D pin setup.
GIS (`/api/v1/gis`)	`GET /dams, GET /defects/spatial, GET /zones/geojson`	PostGIS GeoJSON spatial feature collection endpoint for map rendering.
Digital Twin (`/api/v1/digital-twin`)	`GET /dam/{dam_id}, POST /upload, GET /model/{id}`	3D GLTF model asset retrieval, model activation, defect 3D marker coordinate sync.
Risk (`/api/v1/risk`)	`POST /evaluate/{inspection_id}, GET /matrix/{dam_id}`	Dam Health Index (DHI) calculation, FMEA risk matrix generation, risk rating assignment.
Prediction (`/api/v1/prediction`)	`GET /trend/{defect_id}, GET /rsl/{dam_id}`	XGBoost deterioration trend modeling, growth rate velocity, Remaining Service Life forecast.
Maintenance (`/api/v1/maintenance`)	`GET /work-orders, POST /work-orders, PUT /{id}/complete`	Work order management, priority assignment, repair proof upload, reinspection request.
Reports (`/api/v1/reports`)	`POST /generate, GET /download/{id}, GET /verify-data/{code}`	ReportLab PDF compilation, file streaming, public QR code authenticity verification endpoint.
Alerts (`/api/v1/alerts`)	`GET /, PUT /{id}/read`	System alert notifications for critical defects, overdue maintenance, and risk escalation.
Audit (`/api/v1/audit`)	`GET /logs`	Security audit log querying with user email, action type, and date filters.
Historical (`/api/v1/historical`)	`GET /compare, GET /timeline/{defect_id}`	Multi-inspection spatial defect matching and comparative delta tracking.
Telemetry (`/api/v1/telemetry`)	`GET /live/{dam_id}, POST /log`	Real-time water level, seepage rate, spillway discharge, and sensor log ingestion.
EAP (`/api/v1/eap`)	`GET /plan/{dam_id}, POST /trigger-alert`	Emergency Action Plan (EAP) protocol response matrix and downstream alert trigger.
NDSA (`/api/v1/ndsa`)	`GET /compliance/{dam_id}, GET /checklist`	Statutory NDSA Dam Safety Act 2021 compliance checklist and score calculation.
Supreme (`/api/v1/supreme`)	`POST /directive, GET /directives, PUT /rate-dam`	High-level supreme command directive creation and dam rating assignment (A+ to D).
Public Grievance (`/api/v1/public-grievance`)	`POST /submit, GET /track/{code}, GET /all`	Public grievance submission, citizen tracking code lookup, authority review workflow.

7. AI & MACHINE LEARNING PIPELINE DETAILS

The computer vision and machine learning core consists of three integrated components:

7.1. Deep Learning Defect Detector (YOLO11)

HydroVision AI utilizes an Ultralytics YOLO11 model fine-tuned on custom dam infrastructure inspection datasets. The model detects 6 primary defect classes: *CRACK*, *SPALLING*, *SEEPAGE/EFFLORESCENCE*, *CORROSION*, *EROSION*, and *VEGETATION*. Dynamic confidence filtering rejects detections below 0.40 confidence. Output bounding boxes are stored as normalized coordinates \([x_{\min}, y_{\min}, x_{\max}, y_{\max}]\).

7.2. Computer Vision Measurement Engine (OpenCV)

To provide quantitative millimeter measurements without manual ground measurements, OpenCV computes Ground Sample Distance (GSD) based on flight altitude, camera sensor dimensions, and focal length:

```
GSD (mm/pixel) = ( Altitude (m) × Sensor Width (mm) ) / ( Focal Length (mm) × Image Width (pixels) ) × 1000
```

Using GSD scaling, crack contours are skeletonized via thin-line morphological algorithms to calculate total crack length (mm). Maximum crack width (mm) is determined by computing the maximum Euclidean distance transform value along the defect's medial axis.

7.3. Deterioration Forecasting Model (XGBoost Regressor)

Defect progression velocity \(v = \Delta L / \Delta t\) (mm/year) is modeled using historical inspection series. An XGBoost regressor projects future crack expansion while accounting for environmental severity factors (humidity, freeze-thaw cycles, reservoir head pressure):

```
Remaining Service Life (RSL, Years) = ( Critical Threshold L_max - Current Length L_now ) / ( Growth Velocity v × (1 + α_env) )
```

8. 3D DIGITAL TWIN & GIS SPATIAL ENGINE

HydroVision AI bridges raw GIS coordinates with photogrammetric 3D visualization to allow engineers to inspect defects in spatial context:

- **Three.js WebGL Rendering Engine:** High-resolution `.glb`/`.gltf` photogrammetric dam models are loaded asynchronously with ambient and directional solar lighting. Interactive OrbitControls enable 360-degree rotation, pan, and zoom.
- **3D Defect Pinpointing & Raycasting:** Verified defects contain 3D spatial coordinates \((x, y, z)\). Raycasting click listeners display interactive hover tooltips containing defect code, severity badge, OpenCV measurements, and original drone image thumbnails.
- **Spatial Heatmap Layer:** Defect density heatmaps render on the 3D dam surface to highlight high-stress structural zones requiring urgent grouting or surface treatment.
- **Leaflet GIS Integration:** PostGIS spatial queries stream GeoJSON feature collections to an interactive Leaflet map interface, providing state-wide geographic visualization of dam health and defect spatial clustering.

9. CONDITION SCORING, RISK ENGINE & NDSA GOVERNANCE

The risk engine evaluates dam health according to statutory **NDSA Dam Safety Act 2021** standards using two primary metrics:

9.1. Dam Health Index (DHI) Formula

DHI (0 to 100) represents overall structural health. It starts at 100 and deducts weighted penalties for verified defects based on defect severity and zone criticality:

```
DHI = 100 - SUM( Severity Weight W_s × Zone Weight W_z × Defect Area / Base Area )
```

Zone Weight Multipliers: Spillway (1.5), Upstream Face (1.4), Gallery (1.2), Crest (1.0), Downstream Face (1.1).
Severity Multipliers: Critical (15.0), High (10.0), Medium (5.0), Low (2.0).

9.2. Failure Mode & Effects Analysis (FMEA) Risk Matrix

Risk Score (0 to 100) incorporates structural health, deterioration growth rate, and downstream hazard consequences:

```
Risk Score = ( 100 - DHI ) × 0.60 + ( Growth Velocity Rate Score ) × 0.25 + ( Hazard Consequence Class ) × 0.15
```

Risk Score Range	Risk Level Tag	Condition Rating	Mandatory Statutory Action Required
0.0 – 25.0	**LOW RISK**	GOOD (Grade A)	Routine annual inspection; continue standard monitoring.
25.1 – 50.0	**MODERATE RISK**	WATCH (Grade B)	Bi-annual inspection; log defect growth trends; scheduled maintenance.
50.1 – 75.0	**HIGH RISK**	ATTENTION REQUIRED (Grade C)	Issue priority work order; notify Executive Engineer; detailed non-destructive testing.
75.1 – 100.0	**CRITICAL RISK**	CRITICAL (Grade D)	Emergency Action Plan (EAP) alert; immediate spillway/reservoir drawdown evaluation; Supreme Command notification.

10. REPORTLAB PDF GENERATOR ENGINE ARCHITECTURE

The automated reporting engine (`app/reports/reports.py`) utilizes ReportLab to dynamically build official statutory inspection documents directly from live database records. Key features include:

- **Dual-Language Government Header:** Renders official Hindi and English titles for the Ministry of Jal Shakti and National Dam Safety Authority.
- **Vector Seal Graphics:** Programmatically draws statutory circular vector seals using ReportLab shape primitives (`Drawing`, `Circle`, `String`).
- **Dynamic QR Code Authenticity Check:** Encodes a unique verification URL (`/verify-report?code={report_code}`) into a 75x75 pt QR barcode widget.
- **Public Verification Endpoint:** Anyone scanning the QR code is directed to a public verification page that verifies the SHA-256 report hash against the backend database.
- **Multi-Tier Sign-off Blocks:** Formatted sign-off tables for Senior Inspection Engineer, Executive Engineer, and Chief Authority sign-off.

11. FRONTEND PORTAL & UI ARCHITECTURE

The frontend application is built in React 18 with TypeScript and Vite, structured into a **5 Core Hub Architecture** with strict role guards:

Hub Domain	Routes & Views	Key Features & Functionality
Hub 1: Dam Hub	`Dam Directory (`/dams`), Dam Detail (`/dams/:id`)`	Comprehensive dam inventory, structural metadata, telemetry overview, zone breakdown, supreme rating.
Hub 2: Inspection & AI Studio	`Inspections (`/inspections`), Inspection Detail (`/inspections/:id`), GIS (`/gis`)`	Drone upload studio, AI processing status, human-in-the-loop defect verification, OpenCV measurement editor, spatial GIS map view.
Hub 3: 3D Digital Twin Hub	`Digital Twin (`/digital-twin`), Dam Twin View (`/digital-twin/:damId`)`	Interactive 3D photogrammetric WebGL model view, spatial raycasting, defect marker tooltips, thermal overlay toggle.
Hub 4: Risk & Maintenance Hub	`Maintenance (`/maintenance`)`	Risk matrix overview, maintenance work order creation, contractor assignment, repair evidence upload, reinspection tracking.
Hub 5: Governance & Reports	`Reports (`/reports`), Alerts (`/alerts`), Audit (`/audit`), Supreme Command (`/supreme-command`)`	PDF report generation & download, statutory audit trail log view, emergency command directive issuance, public grievance authority manager.

12. TESTING, VERIFICATION & QUALITY ASSURANCE

System reliability is verified using Pytest via automated test suite `tests/test_phase2c.py`. The suite validates all backend workflow phases:

1. **User Authentication & RBAC Test:** Verifies JWT token generation, password hashing, and role guard enforcement across all 5 user roles.
2. **Dam Asset Creation & Spatial Query Test:** Validates dam CRUD operations and PostGIS geometry column creation.
3. **Drone Upload & Quality Validation Test:** Simulates multi-file upload, EXIF GPS extraction, and image blur scoring.
4. **AI Job Execution & Defect Verification Test:** Mocks YOLO11 bounding box generation and human-in-the-loop engineer modification endpoints.
5. **Risk Engine & DHI Calculation Test:** Asserts exact Dam Health Index scoring and FMEA risk level assignment.
6. **Work Order & Maintenance Lifecycle Test:** Verifies work order creation, progress status updates, and reinspection approval.
7. **ReportLab PDF Generation & QR Verification Test:** Generates a test PDF document, extracts its verification code, and queries `/reports/verify-data/{code}` to confirm 100% data integrity.

13. DEPLOYMENT & EXECUTION PROTOCOL

Local Development Execution:

```
# Backend API Server Setup (Port 8000) cd backend python -m venv venv .\venv\Scripts\activate # On Windows pip install -r requirements.txt python -m app.main # Frontend React SPA Setup (Port 5173) cd frontend npm install npm run dev
```

Docker Production Deployment:

```
docker-compose up -d --build
```

The `docker-compose.yml` orchestrates PostgreSQL 16 with PostGIS, MinIO object storage, FastAPI backend container, and Nginx container serving the React production bundle.

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